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RARE EARTHS: Policy, economic resilience drive new magnet supply chains

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Rare earths, such as those mined at Hastings Technology Metals' Yangibana site in Western

Australia, are in high demand for defense applications and electric vehicles.
Source: Hastings Technology Metals Ltd.

Rare earth elements have come under increasing focus owing to their criticality to defense equipment manufacturing and clean technology. China has monopolized this supply chain for over two decades and has not shied away from using it for geopolitical leverage, forcing others to think of a diversification strategy. This is the first article in a five-part series in which Platts News explores the strategies under play and the challenges they face. Read parts 2, 3, 4, and 5.

Rare earth elements are now indispensable to modern technology, driving advancements across the energy transition, electrification, automation and defense sectors.

As global demand for these critical metals surges, their strategic importance — and the risks of supply concentration — have come into sharp focus. This overview explores the pivotal role of rare earth elements (REE) in manufacturing, their diverse applications, and the evolving landscape of supply chain resilience shaped by government policy and private investment.

REEs comprise 17 metals, including the 15 lanthanides plus scandium and yttrium. They are foundational to the world's most advanced and rapidly growing technologies, spanning energy, digitization and aerospace. In manufacturing, REEs are essential for high-strength permanent magnets, enabling compact, energy-efficient electric motors and generators. These components are critical to electric vehicles, industrial robotics and wind turbines.

Beyond manufacturing, REEs play a vital role in consumer electronics, such as smartphone vibration motors and display color phosphors. They are also integral to energy and environmental technologies, including catalysts, refining processes, and emerging clean-tech supply chains. Defense and aerospace systems rely on REE-based components for precision guidance, sensors, radar, and high-temperature alloys — making supply chain resilience a matter of national security.

The rise of alternative rare earth supply chains

A new wave of rare earth projects is emerging across Australia, Africa, Europe and North & South America with the potential to reshape global trade flows. While refining and processing plans are still emerging, the upstream shift is driven by a strategic imperative among nations and technology manufacturers to diversify REE supply chains away from China.

Key end uses of rare earth elements

Ce: Cerium · Dy: Dysprosium* · Er: Erbium* · Eu: Europium* · Gd: Gadolinium* · Ho: Holmium*
La: Lanthanum · Lu: Lutetium* · Nd: Neodymium · Pr: Praseodymium · Tb: Terbium* · Y: Yttrium*

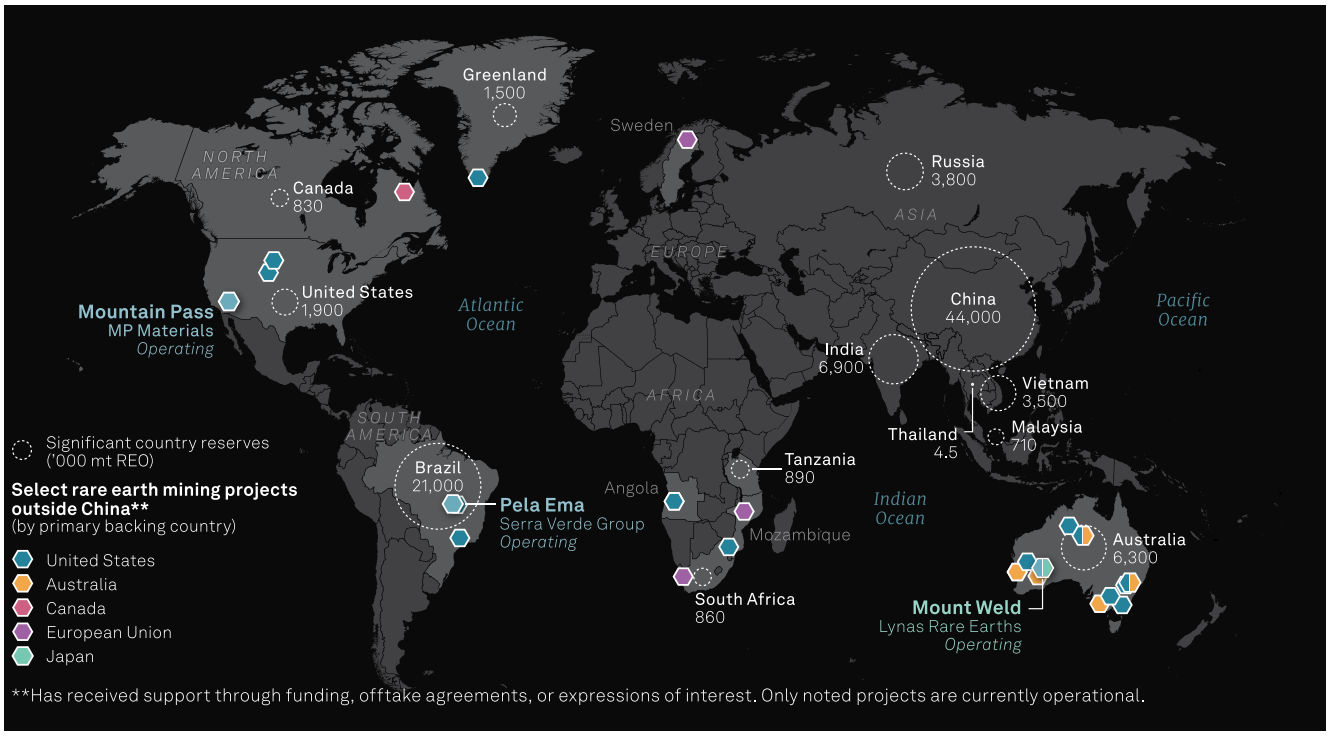
Magnets	Glass	Defense	Metal Alloys	Catalysts	Phosphors	Ceramics
Nd	Nd	Nd	Nd	Nd	Nd	Nd
Tb*	Gd*	Y*	Y*	La	Eu	Y*
Dy*	Er*	Eu	La	Ce	Tb*	Eu
Pr	Ho*	Lu*	Ce	Pr	Y*	Gd*
	Ce	Dy	Pr		Er*	Lu*
	Pr	Sm			Ce	Dy*
		La			Pr	La
		Pr				Ce
						Pr

*Heavy rare earths

Rare earth elements production

Country	2025 ('000 mt)	2024 ('000 mt)
China	270.00	270.00
United States	51.00	45.50
Australia	29.00	29.00
Myanmar	22.00	27.00
Thailand	4.80	2.10
India	2.90	2.90
Madagascar	2.70	1.40
Russia	2.60	2.60
Brazil	2.00	0.56
Nigeria	1.50	1.50
Vietnam	0.15	0.30
Malaysia	0.11	0.14
Other	0.55	1.00

Data compiled from the US Geological Survey (USGS)



S&P Global Energy

Source: S&P Global Energy, US Geological Survey
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Demand growth drivers

Magnetic rare earths — neodymium, praseodymium, dysprosium and terbium — account for roughly 30% of total REE volumes but over 80% of market value, according to McKinsey.

Global demand for magnetic rare earths is expected to triple from 59,000 metric tons in 2022 to 186,000 mt by 2035, McKinsey forecast in a July 2025 report. This growth is driven primarily by automotive applications, projected to expand at a

compound annual growth rate of about 10% between 2025 and 2035. Wind and machinery segments are expected to grow by 5% to 6%, while appliances will see more modest growth of around 3% over the decade.

McKinsey senior partner Michel Van Hoey told Platts, part of S&P Global Energy, that most of this demand centers on dysprosium and terbium, which enable magnets to operate at higher temperatures — a key requirement for hybrid and electric vehicles. Accordingly, McKinsey projects demand for dysprosium and terbium, or DyTb, to grow at a CAGR of about 8.8%, outpacing the 6.1% CAGR forecast for neodymium and praseodymium, or NdPr, which remains the dominant magnet material.

Supply disruptions have prompted some automotive original equipment manufacturers to explore alternative motor designs that use fewer or no REE. However, these alternatives often result in lower power density and increased reliance on other critical minerals such as copper.

Mine-to-magnet supply chain initiatives

McKinsey's findings reflect broader industry trends, with governments and OEMs actively seeking REE supply chains independent of China. China accounts for nearly 60% of global mining, over 85% of processing capacity, and more than 90% of permanent magnet production, according to the US Commerce Department. This makes REEs a strategic vulnerability for governments and industry alike.

In the US, policy has focused on permanent magnets and developing a secure "mine-to-magnet" supply chain, encompassing sourcing, separation, processing, metallization, alloying and magnet manufacturing. The US Defense Department has been especially proactive, investing in established REE and magnet projects through its Manufacturing Capability Expansion and Investment Program.

In July 2025, MP Materials Corp. announced a $400 million public-private partnership with the US Defense Department. The deal included a preferred stock investment and a 10-year agreement granting the department rights to 100% of the permanent magnet output from MP Materials' planned 10X facility.

Expected to begin commissioning in 2028, the plant will boost MP Materials' magnet capacity to around 10,000 mt per year, supplying defense applications such as drones, robotics and advanced semiconductor fabrication.

Australia's Lynas Rare Earths Ltd. has strengthened its US presence through a binding letter of intent with the Defense Department, covering about $96 million in light and heavy rare earth oxide purchases over four years. Lynas also remains a key supplier to Japanese permanent magnet producers and recently extended a long-term offtake agreement with Japan Australia Rare Earths BV, citing new Chinese restrictions on exports of "dual-use items," including REEs, to Japan.

Under the extension, Lynas will supply up to 7,200 mt/year of NdPr to Japan through 2038, and make 75% of its heavy rare earth oxide output available to Japanese customers.

In Brazil, Serra Verde Mineracao Ltda. secured $565 million in financing from the US International Development Finance Corp. to support operations at its Pela Ema deposit in Goiás, where commercial production began in early 2024.

With a product mix rich in dysprosium, terbium and other heavy rare earths, Serra Verde aims to expand output from 5,000 to 6,500 mt/year of total rare earth oxide by end-2027.

New REE and permanent magnet projects are also attracting strong interest from US government agencies. In October 2025, project developer Pensana PLC received a letter of interest from the US Export-Import Bank for up to $160 million in debt funding to support the development of its Longonjo REE mine in Angola.

Production at Longonjo is scheduled to commence in early 2027, with an initial output of 20,000 mt/year of clean mixed rare earth carbonate. A planned second phase could double production to 40,000 mt, representing about 5% of current global rare earth material production.

"OEMs are looking for supply chain resilience — today that means non-China. They want certainty about supply chain provenance at all links, and once they have realigned from China, they want to be able to scale," Paul Atherley, Pensana chairman, told Platts.

Atherley's observations were echoed by REalloys Inc. CEO Lipi Sternheim, who told Platts that demand for rare earths and permanent magnets sourced outside of China is accelerating, driven by government policies and supply-side constraints.

REalloys received a letter of interest from the US Ex-Im Bank in 2025 for a loan of up to $200 million to support the development of its integrated mine-to-magnet North American supply chain. The company's portfolio includes the Hoidas Lake

REE property in Saskatchewan and downstream magnet manufacturing operations in Euclid, Ohio.

"Over the next decade, we expect sustained growth in offtake interest from OEMs, Tier 1 suppliers, and government-linked buyers seeking secure, traceable and geopolitically aligned supply chains," Sternheim said. "From our perspective, this demand is not cyclical — it is policy-driven and supply-constrained."

Platts has proposed four first-of-their-kind price assessments for REEs, focusing on mine-to-industry alternative supply chains. The new assessments launched on March 31.

Rare earth supply chains: Funding, policy and China's edge

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